were followed by a period of relative quiet while the community of physical scientists turned much of its attention to the exploitation of quantum mechanics, as well as to nuclear physics, high-energy particle physics, and advances in observational cosmology.

The concept of black holes had been proposed in a speculative way soon after the discovery of Newton's theory of gravitation. With proper alterations, it was found to have a natural place in the theory of relativity if one was willing to extrapolate solutions of the basic equations to such strong gravitational fields-a procedure which Einstein regarded with skepticism at the time. Using the theory, however, Chandrasekhar pointed out in 1930 that, according to it, stars having a mass above a critical value, the so-called Chandrasekhar limit, should collapse to become what we now call black holes, when they have exhausted the nuclear sources of energy responsible for their high temperature. Somewhat later in the 1930s, this work was expanded by Zwicky and by Oppenheimer and his colleagues, who demonstrated that there is a range of stellar mass in which one would expect the star to collapse instead to a state in which it consists of densely packed neutrons, the so-called neutron star. In either case, the final implosion of the star when its nuclear energy is exhausted should be accompanied by an immense outpouring of energy in a relatively short time, an outpouring to be associated with the brilliance of the supernovae seen occasionally in our own galaxy as well as in more distant nebulas.

World War II interrupted such work. However, in the 1950s and 1960s the scientific community returned to it with renewed interest and vigor on both the experimental and theoretical frontiers. Three major advances were made. First, the knowledge gained from research in nuclear and high-energy physics found a natural place in cosmological theory, providing support for what is commonly termed the "big bang" theory of the formation of our Universe. Many lines of evidence now support the view that our Universe as we know it originated as the result of expansion from a small primordial soup of hot, densely packed particles, commonly called a fireball. The primary event occurred at some time between ten and twenty billion years ago. Perhaps the most dramatic support for the hypothesis was the discovery of the degraded remnants of the light waves that accompanied a late phase of the initial explosion.

Second, the neutron stars predicted by Zwicky and the Oppenheimer team were actually observed and behaved much as the theory predicted,